

Precept 7

Goals

- Connections among the cumulative distribution function, probability density function, cumulative hazard function, and hazard rate function.
- The formula $\mathbb{E}[L] = \int_0^\infty \mathbb{P}\{L > t\}dt$ for a continuous and positive random variable L .
- Manipulation of Poisson events.
- Connection between exponential and gamma distributions and Poisson arrival times.
- Application of superposition and thinning principles.

Reminder

Functions of lifetime (F, f, H, h) : Let L be the lifetime of some device. Assume that L is a continuous random variable and $0 < L < \infty$. The cumulative distribution function of L is defined as $F(t) = \mathbb{P}\{L \leq t\}$. Given the cumulative distribution function of L , one can find the probability density function by $f(t) = \frac{dF(t)}{dt}$. Note that one expects $\mathbb{P}\{L > t\}$ to decrease with respect to t . Furthermore, $\mathbb{P}\{L > 0\} = 1$, and $\lim_{t \rightarrow \infty} \mathbb{P}\{L > t\} = 0$ as we have $0 < L < \infty$. This is the intuition why the lifetimes are modeled as

$$\mathbb{P}\{L > t\} = e^{-H(t)},$$

where $H(t) (= -\log(1 - F(t)))$ is an increasing function, namely "cumulative hazard function". Furthermore, H satisfies $H(t) = \int_0^t h(s)ds$, for some positive function h , namely "hazard rate". Note that we have $\int_0^\infty h(t)dt = \infty$ so that $\lim_{t \rightarrow \infty} \mathbb{P}\{L > t\} = \lim_{t \rightarrow \infty} e^{-\int_0^t h(s)ds} = 0$. Given any of these functions, one can immediately calculate all the others, e.g.,

$$F(t) = \mathbb{P}\{L \leq t\} = 1 - e^{-H(t)} = 1 - e^{-\int_0^t h(s)ds}$$
$$f(t) = \frac{dF(t)}{dt}, \quad h(t) = \frac{dH(t)}{dt}, \quad H(t) = -\log(1 - F(t))$$

Formula for calculating expectations: Let L be a continuous and **positive** random variable. Then

$$\mathbb{E}[L] = \int_0^\infty \mathbb{P}\{L > t\}dt.$$

Poisson processes:

Definition 1. A random process $\{N_t\}_{t \geq 0}$ is called a *counting process* if:

1. $N_0 = 0$

2. N_t is increasing in t
3. N_t increases by one every time that it changes

Definition 2. A counting process $\{N_t\}_{t \geq 0}$ is called a Poisson process with rate $\lambda > 0$ if :

1. $N_t - N_s \perp\!\!\!\perp \{N_r\}_{r \leq s}$ for $t \geq s$ (independent increments)
2. $N_t - N_s \sim N_{t-s} \sim \mathcal{P}(\lambda(t-s))$ for $t \geq s$ (stationary increments)

Arrival times Let $\{N_t\}_{t \geq 0}$ be a Poisson process with rate $\lambda > 0$ and let T_k denote the time at which the k -th arrival occurs. Let us consider the first arrival time T_1 .

$$\mathbb{P}(T_1 > t) = \mathbb{P}(N_t = 0) = \frac{e^{-\lambda t}(\lambda t)^0}{0!} = e^{-\lambda t} \quad (1)$$

So $T_1 \sim \mathcal{E}(\lambda)$. We also have that $T_1, T_2 - T_1, \dots, T_k - T_{k-1}$ are i.i.d. according to an exponential distribution with parameter λ , so $T_k = T_1 + (T_2 - T_1) + \dots + (T_k - T_{k-1}) \sim \Gamma(k, \lambda)$.

Theorem 3. (*Principle of Superposition*) If $N_t^1, \dots, N_t^k$ are independent Poisson processes with rates $\mu_1, \dots, \mu_k$, then $N_t^1 + \dots + N_t^k$ is a Poisson process with rate $\mu_1 + \dots + \mu_k$.

Theorem 4. (*Principle of Thinning*) If N_t is a Poisson process with rate μ , and if each arrival is independently assigned one of k labels with probabilities $p_1, \dots, p_k$, then the counting processes $N_t^1, \dots, N_t^k$ that count the arrivals of each type are independent Poisson processes with rates $p_1\mu, \dots, p_k\mu$.

Exercises

Exercise 1 (Eggs gone bad)

"Prince-Udon" has been making waves in the food critics circles lately and the famed celebrity chef Jordan Clamsey decides to pay a surprise visit. Jerry, the owner of the above-mentioned restaurant, is clearly flattered by such attention and is planning to impress the critic with a classic ramen with an egg. However, he shortly realizes in agony that each of the n eggs in the fridge will go bad uniformly at random at some point within an hour. Given that all eggs' expiry times are independent, what is the distribution of the time when Jerry is able to find at least one good egg? Find expected value of this time and evaluate the failure rate.

Exercise 2 (Basics of Poisson processes and conditional distribution of arrival times)

Consider the following situation:

- A very dedicated ORF309 TA is answering last-minute questions via email from his/her dorm room the night before the exam, starting at 10 p.m., or $t = 0$.
- Equally dedicated ORF309 students are studying all night for the exam, starting at 10 pm ($t = 0$) until 10 am ($t = a$). They send emails to the TAs according to a Poisson process with rate λ .
- After each email, the TA decides to either stay up and wait for additional questions or go to bed.
- The TA considers the night a success if he/she goes to bed and no further questions arrive. The night is a failure if he/she goes to bed and another question does arrive OR the TA does not go to bed and no further questions arrive (in this case, the TA lost sleep for no reason).
- The TA decides to use the following policy: fix $s \in [0, a]$ beforehand and go to bed after the first email that arrives after s .

What is the probability of a successful teaching night for a fixed s ? What is the value of s that maximizes this probability and what is this optimal probability? Given that the night is a success, what is the distribution of the amount of sleep the ORF309 TA gets under policy s , assuming the alarm is set for $t = a$?

Exercise 3 (Poisson superposition and Bayes' theorem)

In a car repair shop wheels of the cars are inspected for safety and a repair/change is made in case it is necessary. The arrival process of cars requiring repairs of i wheels are independent Poisson processes with respective rates λ_i , where $\lambda_0 = 4, \lambda_1 = 2, \lambda_2 = 3, \lambda_3 = 2, \lambda_4 = 4$. The inspection time and the repair time of each wheel are both exponentially distributed random variables with mean 20 minutes. We can assume that the inspection time, and the repair times of the wheels are independent of each other and of the customer arrival process. On a given day, it took 50 minutes to complete the job for the first car arrived to the shop. What is the probability that 2 wheels of that car needed a repair?